

Portfolio

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Nergiz

Professional Experience

Valeo Telematik und Akustik GmbH

SEP 2019 – AUG 2025 , FRIEDRICHSDORF - GERMANY



Project 1 – Software Architecture (Sep 2019 – Aug 2025)

Development of the software component state machine for the telematics control unit (TCU).

MAIN TASKS:

- Analysis of customer requirements on IBM Doors.
- Modeling of architecture diagrams as well as state machine with UML elements on Enterprise Architect.
- Code generation (C++) from state machine diagrams with Lieber Lieber Embedded Engineer.
- Collaboration with C++ developers to combine the developed code with generated code.

APPLIED METHODS & TOOLS

Enterprise Architect

Lieber Lieber Embedded Engineer

CoreALM

JIRA

Gerrit (Git)

IBM Doors

Eclipse

SomelP

Yocto Project

C++

Valeo Telematik und Akustik GmbH

SEP 2019 – AUG 2025 , FRIEDRICHSDORF - GERMANY



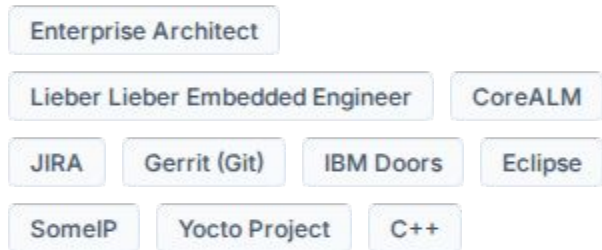
Project 2 – Software Architecture (Sep 2019 – Aug 2025)

Development of software architecture of the Valeo Platform software component Positioning Manager for the telematics control unit (TCU).

MAIN TASKS:

- Creation and management of requirements on IBM Doors.
- Modeling of architecture diagrams with UML elements on Enterprise Architect.
- Analysis of customer requirements for positioning functions to update the Valeo platform requirements when a new functionality should be part of the platform.
- Collaboration with C++ developers for code development based on defined requirements.

APPLIED METHODS & TOOLS



Continental Teves AG & Co. oHG (via FERCHAU)



AUG 2018 - JUL 2019 , FRANKFURT AM MAIN - GERMANY

Project 1 – Software Architecture (Aug 2018 – Jul 2019)

Support of architecture-upgrades to build a direct communication between AUTOSAR-based Software Components and the Software Components, which are generated via cIDL (Continental's Interface Description Language).

MAIN TASKS:

- Analysis of AUTOSAR interfaces for data exchange (sender-receiver) as well as function call (client-server) between components.
- Analysis of cIDL interfaces for data exchange as well as function call between components together with scheduling.
- Development of several solution options to establish direct communication between AUTOSAR and cIDL components.

APPLIED METHODS & TOOLS

Rhapsody

Continental CESSAR-CT

Eclipse

GitHub

C

Continental Teves AG & Co. oHG (via FERCHAU)



AUG 2018 - JUL 2019 , FRANKFURT AM MAIN - GERMANY

Project 2 – Multi-Core Software (Aug 2018 – Jul 2019)

Preparation of introduction documentation as well as instructions for use for Atomic Operations Library of programming language C together with application examples over existing product software.

MAIN TASKS:

- Analysis of Atomic Operations provided by C programming language since 2011 standard as well as Atomic Library of Elektrobit Tresos OS.
- Analysis of advantages of Atomic Operations against MUTEX based algorithms.
- Preparation of application examples with existing product software (e.g. manually developed Proxy (master-slave) for intra-core communication between AUTOSAR BSW components).

APPLIED METHODS & TOOLS

Eclipse

GitHub

GNU Compiler

C

FEV GmbH (Resident at BMW)

JAN 2017 - JULY 2018 , MUNICH - GERMANY



Project 1 – Model-based Software/System Modeling & Architecture (Apr 2017 – May 2018)

SysML based modeling of software component "Radmomentstellen" with BMW-specific methodology "SMArDT" (Specification Method for Architecture Design and Test). Creation of mapping between requirements and SysML artifacts.

MAIN TASKS:

- Analysis of requirements regarding software component "Radmoment Stellen".
- Modeling of use case diagrams, activity diagrams and internal block diagrams.
- Creation of mapping between requirements and SysML artifacts.
- Creation of new requirements if necessary.

APPLIED METHODS & TOOLS

IBM Doors

PTC Modeler

FEV GmbH (Resident at BMW)

JAN 2017 - JULY 2018 , MUNICH - GERMANY



Project 2 – ECU Test and Simulation Setup (Restbus) (Jan 2017 – Mar 2017)

Development of a residual bus simulation setup for an ECU for differential lock control, where Vector CANoe tool and CAPL Script were used.

MAIN TASKS:

- Analysis of CAN and Flex-Ray vehicle networks and preparation of the signal list of regulated limited slip differential ECU.
- Editing of CAN and Flex-Ray simulation models and development of CRC (Cyclic Redundancy Check) algorithms with CAPL scripting language.
- Testing of communication for the necessary signals.

APPLIED METHODS & TOOLS

Vector CANoe

CAPL Script

ECU-Test

FEV GmbH (Resident at BMW)

JAN 2017 - JULY 2018 , MUNICH - GERMANY



Project 3 – Test Case Creation for Vehicle Application Software (Jun 2018 – Jul 2018)

Test case creation for terminal control (Park-Live-Drive / Parken-Wohnen-Fahren) functions.

MAIN TASKS:

- Analysis of specifications of terminal control functions.
- Creation of test cases based on requirements and success criteria.

APPLIED METHODS & TOOLS

HP ALM

JIRA

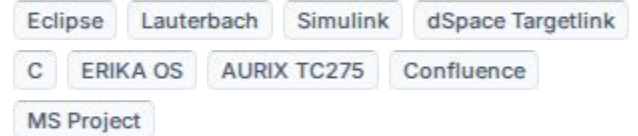
Project 1 – AMALTHEA4PUBLIC Research Project (Jul 2015 – Dec 2016)

Technical project management and realization of research project AMALTHEA4PUBLIC.

MAIN TASKS:

- Creation of multi-core software application scenario based on existing control software for an electric vehicle with range extender.
- Creation of software and hardware models on Eclipse-based AMALTHEA tool of Robert Bosch GmbH in accordance with existing software components and hardware specifications of Infineon Aurix TC275 microcontroller based platform.
- Configuration of ERIKA OS operating system for Infineon Aurix TC 275 based HW platform.
- Integration of dSpace TargetLink generated vehicle control software on HW platform as 2-cores application.
- Management of technical project tasks as well as continuous communication between AVL and other work package partner companies as work package leader of "Demonstration" work package.

APPLIED METHODS & TOOLS



Project 2 – Tool Development for Static Code Analysis (Jul 2015 – Dec 2015)

Support for AVL England branch in a transmission software development project on the topic of static code analysis. Development of a software tool with useful human-machine interface for further analysis of .txt file-based results of the PC-Lint tool.

MAIN TASKS:

- Analysis of PC-Lint API and structure of files as static code analysis results.
- Modeling of human-machine interface for SW tool based on requirements.
- Development of SW tool with C# programming language and verification against requirements.

APPLIED METHODS & TOOLS

PC Lint

Visual Studio

C#

PTC Integrity

MISRA-C

Project 3 – Tool Development for Model Analysis (Jan 2016 – Apr 2016)

Support for AVL England branch in a transmission software development project on the topic of SW model analysis. Development of a MATLAB-based software tool with useful human-machine interface for multiple analysis of Simulink models against MISRA and AVL model-based development standards.

MAIN TASKS:

- Analysis of MXAM API for MATLAB Script based development.
- Development of SW tool and interface with MATLAB Script programming language.
- Testing of SW tool.

APPLIED METHODS & TOOLS

MXAM (Model Examiner)

Simulink

MATLAB Script

MISRA Standard

PhD Dissertation – Validation Platform

Desktop Microfactory Concept System Automation

Software design and workflow synchronization to automate micrometer-precision manipulation, laser marking, and visual inspection.

MAIN TASKS:

- Designed and developed the software for the Desktop Microfactory Concept including the human-machine interface (with C# .Net Framework) and the embedded real-time software (C code) that runs on the dSpace DS1005 RCP/HIL PowerPC System.
- Built and coordinated parallel real-time blocks for multiple subsystems, including a DC Rotational conveyor, two 3-DOF Delta assembly robots, and a galvo laser scanner.
- Developed automated autofocus algorithms using a miniature digital microscope and active focus mechanics.
- Engineered a central finite state machine coordinating: sample delivery, Delta-robot vacuum pick-place, galvo laser contour marking, and visual template matching.
- Successfully marked a 550 μm "G" letter contour on 3 mm spherical steel balls and automatically sorted them based on visual quality indicators.

APPLIED METHODS & TOOLS

dSPACE DS1005

Delta Robot Kinematics

Galvo Scanner Calibration

TIMM 400 Microscope

Template Matching Vision

Finite State Automation

Vacuum Gripper

Micro-Manipulation

C

C# .NET Framework

SPI Pulsed Fiber Laser

Analog Optical Encoders